

Chapter 1 — Exercises

Multiple-Choice Questions

Test your understanding of the Introduction

Q1. What is machine learning best described as?

- A. Storing large amounts of data efficiently
- B. Extracting knowledge and patterns from data
- C. Writing explicit rules for every possible situation
- D. Manually labeling every data point by hand

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- B. Extracting knowledge and patterns from data**
- C. Writing explicit rules for every possible situation
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Answer: **B** — Machine learning is about extracting knowledge and patterns from data, rather than hand-coding rules.

Q2. Which of these is a *supervised* learning task?

- A. Grouping customers into segments not known in advance
- B. Discovering unknown topics in a set of blog posts
- C. Predicting whether a tumor is benign or malignant from labeled images
- D. Flagging abnormal website access with no examples of "abnormal"

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Answer: **C** — Supervised learning requires labeled input/output pairs, such as images labeled as benign or malignant.

Q3. Why do handcoded rules fail for tasks like face detection?

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- B. They require a GPU to execute
- C. They can only be written in Python
- D. Humans can't express the decision as explicit rules over pixels

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D. Humans can't express the decision as explicit rules over pixels

Answer: **D** — Humans cannot write explicit rules for pixel-level decisions like face detection, so we rely on machine learning to learn from examples.

Q4. In a data table, a row and a column represent:

- A. a feature and a sample, respectively
- B. a sample and a feature, respectively
- C. a label and a model, respectively
- D. a parameter and a target, respectively

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Answer: **B** — In a data table, rows correspond to samples (data points), and columns correspond to features (attributes of the data).

Q5. Why do we split data into training and test sets?

- A. To estimate how well the model generalizes to unseen data
- B. To make the model train faster
- C. To reduce the number of features
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- A. To estimate how well the model generalizes to unseen data**
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- C. To reduce the number of features
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Answer: **A** — The test set is used to evaluate the model's performance on unseen data, providing an estimate of its generalization ability.

Q6. Classifying an iris as setosa / versicolor / virginica is:

- A. a regression problem
- B. a binary classification problem
- C. a multiclass classification problem
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Answer: **C** — Since there are three possible classes (setosa, versicolor, virginica), this is a multiclass classification problem.

Q7. The common scikit-learn workflow for a model is:

- A. load → save → print
- B. fit → predict → score
- C. map → reduce → filter
- D. compile → run → debug

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Answer: **B** — The typical workflow in scikit-learn involves fitting the model to training data, making predictions on new data, and scoring the model's performance.

Q8. With `n_neighbors=1`, k-NN predicts a new point as:

- A. the class of the single nearest training point
- B. the average of all training labels
- C. the majority class of the whole dataset
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Answer: **A** — In 1-NN (k-nearest neighbors with $k=1$), the predicted class for a new point is simply the class of its closest training point.

Q9. Which library provides the DataFrame for tabular data?

- A. NumPy
- B. SciPy
- C. pandas
- D. matplotlib

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Answer: **C** — The pandas library provides the DataFrame structure, which is ideal for handling tabular data, while NumPy provides arrays.

Q10. A defining feature of *unsupervised* learning is:

- A. it always achieves higher accuracy
- B. it needs labeled input/output pairs
- C. it requires a separate test set in order to train
- D. it has no known outputs and finds structure in the data

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Answer: **D** — Unsupervised learning does not rely on labeled outputs; instead, it seeks to find patterns or structure in the input data.

Q11. `load_iris()` returns the dataset as:

- A. a pandas DataFrame
- B. a `Bunch` object (dictionary-like)
- C. a plain Python list
- D. a CSV file on disk

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Answer: **B** — The `load_iris()` function from scikit-learn returns the dataset as a `Bunch` object, which is similar to a dictionary and contains the data, target labels, and other metadata.

Q12. For a classifier, `knn.score(X_test, y_test)` returns:

- A. the number of neighbors used
- B. the predicted class labels
- C. the accuracy on the test set
- D. the training-set error

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Answer: **C** — The `score` method for a classifier in scikit-learn computes the accuracy of the model on the provided test data and labels.

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	B	ML <i>extracts knowledge from data</i> instead of hand-coding rules
2	C	Labeled input→output pairs make it supervised
3	D	No one can write pixel-level rules for a face → learn from examples
4	B	Data is samples × features — rows are samples, columns features
5	A	The test set measures generalization to unseen data
6	C	Three possible classes → multiclass classification

Answers 7–12

Q	Ans	Why
7	B	Every estimator follows <code>fit</code> → <code>predict</code> → <code>score</code>
8	A	1-NN copies the label of the single closest training point
9	C	pandas provides <code>DataFrame</code> ; NumPy provides arrays
10	D	Unsupervised learning has no labels — it finds structure
11	B	<code>load_iris()</code> returns a Bunch (dict-like) with <code>data</code> , <code>target</code> , ...
12	C	For a classifier, <code>score</code> returns accuracy on the given data

Well done!

Review Chapter 1 for any you missed